**Table 11.1 When and where planning is employed and used in various applications**

| Application | Implemented | Environment | Purpose | Timing | Configuration |
|---|---|---|---|---|---|
| Personal assistant | At or within the LLM | Personal device | Facilitate tool use | During the response | As part of the prompt or LLM |
| Customer service bot | Not typical; restricted environment | Restricted environment, no tool use | | | |
| Autonomous agent | As part of the agent prompt and within the LLM | Server or service | Facilitate complex tool use and task planning | As part of constructing the agent and/or during the response | Within the agent or LLM |
| Collaborative workflows | As part of the LLM | Shared canvas or coding | Facilitate complex tool use | During the response | Within the LLM |
| Game AI | As part of the LLM | Server or application | Complex tool use and planning | Before or during the response | Within the LLM |
| Research | Anywhere | Server | Facilitate tool use and engage in complex task workflows | Before, during, and after response generation | Anywhere |

Table 11.1 shows several varied application scenarios in which we may find an assistant or agent deployed to assist in some capacity. To provide further information and guidance, this list provides more details about how planning may be employed in each application:

- *Personal assistant*—While this application has been slow to roll out, LLM personal assistants promise to surpass Alexa and Siri in the future. Planning will be essential to these new assistants/agents to coordinate numerous complex tasks and execute tools (actions) in series or parallel.
- *Customer service bot*—Due to the controlled nature of this environment, it’s unlikely that assistants engaged directly with customers will have controlled and very specific tools use. This means that these types of assistants will likely not require extensive planning.